

Plate Loading Challenge

Problem ID: problem-36-24527669

The University Gym is running a plate-loading challenge. Before each attempt, the instructor present announces a target weight. Your goal is to load the heaviest barbell that does not exceed the announced target. The equipment officer also wants to know how many different selections of plate pairs could produce that desired weight.

The empty bar alone weighs B units. The rack holds n labelled pairs of plates, numbered 1 to n . Pair i contains two plates, each weighing p_i units. To use a pair, load each end of the bar with the plates, adding $2p_i$ units in total. Each pair may only be used at most once, and it cannot be split up or combined with another. There is also enough room on the bar for any number of selections.

2 selections are different if they use different sets of pair labels. Pairs with equal weights still have different labels. Rearranging the selected plates does not create a different selection. Selecting no pairs is allowed and it also counts as 1 selection.

All weights are expressed in integer units, where one unit is = 100 grams. The target T includes the weight of the bar. Find the maximum achievable total weight at most T , and the exact number of selections achieving it.

Input

The first line contains three integers n , B , and T , where $1 \leq n \leq 200$, $1 \leq B \leq 1000$, and $B \leq T \leq B+10000$.

The second line contains n integers $p_1, \dots, p_n$, where $1 \leq p_i \leq 1000$. The i th integer is the weight of *one plate* from pair i , in the same units as B and T . There is one challenge per input file.

Output

Output two integers on one line, separated by a space: the maximum total barbell weight in the given units, and the exact number of selections achieving that weight. The count may exceed the range of a 64-bit integer.

Sample explanations

In Sample 1, the empty bar weighs 200 units (20 kg). The selections $\{1, 2\}$, $\{1, 3\}$, and $\{2, 3, 4\}$ each add 160 units. They are the three selections achieving the target of 360 units.

In Sample 2, pair 1 alone gives 212 units and pair 2 alone gives 220 units. Both pairs together would exceed the target of 221 units, so the answer is 220 units with one selection.

In Sample 3, the target equals the empty bar's weight. The only possible selection is the empty selection.

Sample Input 1	Sample Output 1
4 200 360 50 30 30 20	360 3
Sample Input 2	Sample Output 2
2 200 221 6 10	220 1
Sample Input 3	Sample Output 3
3 200 200 10 20 30	200 1

D-ary Codes

Problem ID: problem-36-23810308

You have unearthed a repository of classic texts from the planet Earth. You are certain these texts have great archeological (and financial) value. Unfortunately, the computer is bulky and you cannot take it with you. The locals outlawed all digital technology after the AI Apocalypse and destroy computers on sight. You have smuggled in a few flash drives, but your storage capacity is limited.

Moreover, the source text was encoded using binary 0s and 1s, while your black-market storage devices use *many-valued logic*. For example, your ternary drive stores data using ternary digits, a.k.a ‘trits’, where each digit can be 0, 1 or 2.

Can you find the optimal variable-length code for compressing a given text using d distinct symbols?

Input

The first line contains two integers N and d , where N is the number of lines in the text, and d is the number of symbols used by your storage drive. Each of the next N lines contains between 2 and 80 Unicode characters, where the length includes the line ending `\n`. The tab character `\t` will never appear.

Problem bounds are $1 \leq N \leq 10^5$ and $3 \leq d \leq 10$. The maximum number of characters is 80×10^5 . Additionally, the number of unique characters k in any given text, including `\n`, is bounded by $2 \leq k \leq 100$.

Important: New lines `\n` are part of the text and should be included in the encoding. On Windows, line endings are represented by two characters, `\r\n`. To avoid any operating system-specific issues, only count line feed characters `\n`. Any carriage return characters `\r` should be ignored.

Output

Output the length of the encoded text using an optimal variable-length code for an alphabet of size d .

Sample Input 1

```
1 3
Fair is foul, and foul is fair.
```

Sample Output 1

```
76
```

Sample Input 2

```
2 8
To-morrow, and to-morrow, and to-morrow,
Creeps in this petty pace from day to day
```

Sample Output 2

```
120
```

Radio network

Problem ID : problem-36-22913428

Time limit : 3s

Telstra plans to build a new radio network connecting n stations across Perth. There are m possible fibre links between pairs of stations. Each link has a construction cost between 1 and c . Telstra wants all stations to be connected, directly or indirectly, while minimising the total construction cost.

Determine the smallest total construction cost that allows messages to travel between any pair of stations in the finished network.

Input

The first line contains three integers n , m and C , ($1 \leq n \leq 10^5$, $0 \leq m \leq 10^5$, $1 \leq C \leq 100$) representing the number of stations, the number of possible fibre links and the maximum construction cost.

Each of the next m lines contains three integers u , v and w , ($1 \leq u, v \leq n$, $u \neq v$, $1 \leq w \leq C$) representing a possible fibre link between stations u and v with construction cost w . The stations are numbered from 1 to n . Each fibre link allows communication in both directions. Multiple fibre links may exist between the same pair of stations.

Output

Output the minimum total construction cost that allows messages to travel between every pair of stations. If this is impossible, output -1.

Sample input 1

```
5 7 5
1 2 3
1 3 1
2 3 2
2 4 4
3 4 3
3 5 5
4 5 2
```

Sample output 1

```
8
```

Sample input 2

```
4 2 5
1 2 1
3 4 2
```

Sample output 2

```
-1
```

Race to the Beginning

Problem ID: problem-36-23668796

In the far future, scientists have built a network of time stations. Each station may contain one-way time machines which transport a traveller from one station to another.

Every time machine changes the traveller's position in time by some integer amount. A positive value moves the traveller forward in time, while a negative value moves the traveller backwards in time.

You begin at time station 0.

You may travel through the time machines as many times as you want, as long as the direction of each machine is followed.

Your goal is to determine whether it is possible to travel arbitrarily far backwards in time. In other words, for any point in the past, determine whether there is some sequence of time-machine journeys starting from station 0 which can take you even further into the past.

Input

The first line contains two integers N and M , where N is the number of time stations and M is the number of one-way time machines.

The stations are numbered from 0 to $N - 1$.

Each of the next M lines contains three integers u , v , and t .

This means there is a one-way time machine from station u to station v which changes time by t units.

A positive value of t moves the traveller forward in time, while a negative value moves the traveller backwards in time.

The constraints are:

$$1 \leq N \leq 700$$

$$0 \leq M \leq 5000$$

$$0 \leq u, v < N$$

$$-10^6 \leq t \leq 10^6$$

There may be multiple time machines between the same pair of stations.

Output

Output YES if, starting from station 0, it is possible to travel arbitrarily far backwards in time.

Otherwise, output NO.

Sample Input 1

```
4 4
0 1 3
1 2 -5
2 1 2
2 3 4
```

Sample Output 1

YES

Sample Input 2

```
3 3
0 1 4
1 2 -2
2 1 3
```

Sample Output 2

NO

Sample Input 3

```
5 3
0 1 2
3 4 -5
4 3 1
```

Sample Output 3

```
NO
```

Chrono Delivery

Problem ID : problem-36-22971029
Time limit : 3s

The Chrono Delivery Guild ships parcels between the N space stations of the Outer Reach, which are joined by one-way wormholes, each with a travel time. Some are *temporal* wormholes with negative travel time. The dispatcher Spock must find the minimum total time from station a to station b for each delivery request, using wormholes any number of times. If a route can pass through a *time loop*, a cycle with negative total time, the courier could arrive arbitrarily early, and the request is a paradox.

Input

The first line contains three integers N , M and Q ($1 \leq N \leq 150$, $0 \leq M \leq 10\,000$, $1 \leq Q \leq 100\,000$): the number of stations, the number of wormholes and the number of delivery requests. Stations are numbered 0 to $N - 1$.

Each of the next M lines contains three integers u , v and t ($0 \leq u, v < N$, $-10\,000 \leq t \leq 10\,000$), describing a one-way wormhole from station u to station v with travel time t . There may be several wormholes between the same pair of stations, and a wormhole may lead from a station back to itself.

Each of the next Q lines contains two integers a and b ($0 \leq a, b < N$): a request to travel from station a to station b .

Output

For each request, in the order given, output a single line containing:

- UNREACHABLE if there is no route from a to b ;
- PARADOX if some route from a to b passes through a station that lies on a time loop, so arbitrarily small times are possible;
- otherwise, the minimum total travel time from a to b as an integer (which may be negative).

A courier who does not move takes time 0, so a request with $a = b$ has answer 0 unless it is a paradox.

Note

With up to 100 000 requests, reading input line by line with `input()` may be slow in Python. Consider reading all input at once with `sys.stdin.buffer.read().split()`.

Sample input 1

```
4 4 4
0 1 -2
0 2 1
2 3 -3
3 2 2
0 1
0 3
3 1
1 1
```

Sample output 1

```
-2
PARADOX
UNREACHABLE
0
```

Sample input 2

```
5 5 5
0 1 4
1 2 -2
2 1 1
2 3 3
0 4 7
0 3
0 4
1 1
3 1
0 0
```

Sample output 2

```
PARADOX
7
PARADOX
UNREACHABLE
0
```